Syllabus Details

Syllabus ID:	14178
Syllabus Name:	**Software Testing_Kiểm thử phần mềm**
Course Name English:	**Software Testing**
Subject Code:	**SWT301**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h (60 sessions) contact hours + 145-minute final exam + 102,6h self-study
Pre-Requisite:	SWE102 or SWE201c or SWE202c
Description:	• General principles of software testing, including its purposes and objectives. • Testing integrated into software development processes, covering test levels, test types, and maintenance testing. • Static testing basics, including feedback and review processes to enhance software quality. • Techniques such as black-box (functional), white-box (structural), and experience-based testing applied to derive test cases. • Test activities like planning, estimation, monitoring, risk management, defect reporting, and test automation usage. • Tools and methods to support effective testing, emphasizing benefits and potential risks. • AI is used to explore testing concepts, support static testing, and create/optimize Test Plans.
StudentTasks:	Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Students are responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time. - The use of AI in assessments and exams must comply with the instructions of the course instructor. - Use laptop in class only for learning purpose - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Unit Test Framework - Bug Management Tool - Ms Office - IDE - Static Analyst Tool - Automation Tool - AI Tools: Grok (xAI), ChatGPT, Gemini, Deepseek
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	In the case: (5 > Final TE Score >=4) & (5 > Final PE Score >=4) & FR < 5, the student can choose to take the resit of both TE & PE OR just either TE or PE.
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online
1	Foundations of Software Testing: ISTQB Certification	Dorothy Graham, Erik van Veenendaal, Rex Black	Cengage Learning, EMEA	2020	4th	978-1-4737-6479-8	True	True	False
2	Agile Testing: A Practical Guide for Testers and Agile Teams	Lisa Crispin,Janet Gregory				978-0321534460	False	True	False

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online
3	Computer Software Validation. Alan Muirhead. News & Views, March 2000. Available at http://www.stcpmc.org/archive/archive/n&v/feat0300.html						False	False	True
4	Course Software Testing and Quality Assurance by Tian in http://engr.smu.edu/~tian/class/7314.06f/syl.html (availble in FU library)						False	False	True

10 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Know definitions, concepts and terminologies about Software Testing, using AI to explore testing principles and real-world testing scenarios through effective prompts.
2	CLO2	Understand the software testing process. Distinguish between levels of testing (e.g. unit, integration, system, acceptance) and types of testing (e.g. functional, non-functional) applied to a specific group project (encouraged to choose project in SWP391)
3	CLO3	Use static testing techniques to detect errors through a review process, using AI to review documents for defects (e.g., ambiguity, inconsistency) and analyze static code for errors (e.g., code smells, bug patterns).
4	CLO4	Use dynamic testing techniques like black-box, white-box, and experience-based testing to design and execute test cases, and determine test coverage.
5	CLO5	Develop a comprehensive test plan for an AI-enabled project, clearly defining its scope, goals, resources, and testing schedule.
6	CLO6	Analyze and prioritize risks to focus testing efforts effectively. Understand and apply bug logging, bug reporting, and test reporting practices.
7	CLO7	Classify and select appropriate testing tools, evaluate the benefits and risks of test automation, and choose an automation tool for testing functionalities in the team project.
8	CLO8	Understand and apply Agile testing methods.
9	CLO9	Utilize AI tools to support learning and exercises related to the course.
10	CLO10	Demonstrate presentation skills and effective teamwork

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	- Course Introduction - Introducing AI integration in teaching and learning, as well as assessment	Offline	LO1 - LO9		Slides, AI tool	SWT301	- Divide into groups - Discuss and ask questions about the assignment and assessment	
2	Topic 1. Fundamentals of testing 1.1 What is testing?	Offline	LO1		text book: Chapter 1 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
3	1.2 Why is testing necessary?	Offline	LO1		text book: Chapter 1 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
4	1.3 Testing principles	Offline	LO1		text book: Chapter 1 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
5	1.4 Fundamental test process	Offline	LO1		text book: Chapter 1 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
6	1.5 The psychology of testing 1.6 Codes of ethics	Offline	LO1		text book: Chapter 1 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
7	Guiding Lab 1 (Use chat AI to explore testing fundamentals through effective prompts)	Offline	LO1, LO9		text book: Chapter 1, AI Tool		Teamwork and Self Study - Read all relevant materials - Use Chat AI - Self Practice	
8	Topic 2. Testing throughout the software life cycle 2.1 Software development models	Offline	LO2		text book: Chapter 2 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
9	2.2 Test levels	Offline	LO2		text book: Chapter 2 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
10	2.2 Test levels (cont.)	Offline	LO2		text book: Chapter 2 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
11	2.3 Test types: the targets of testing	Offline	LO2		text book: Chapter 2 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
12	2.4 Maintenance testing	Offline	LO2		text book: Chapter 2 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
13	Progress test 1	Offline	LO1, LO2		text book: Chapter 1 and Chapter 2		- Enter the exam system - Complete Progress Test	
14	Review Lab 1	Offline	LO1, LO9		text book: Chapter 1, AI Tool		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
15	Guiding Lab 2 (Install and study an unit test framework, Use IDE to write unit test code and run unit test)	Offline	LO2		text book: Chapter 2, AI Tool		Teamwork and Self Study - Read all relevant materials - Use Chat AI - Self Practice	
16	Topic 3. Static techniques 3.1 Reviews and the test process	Offline	LO3		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
17	Topic 3. Static techniques 3.2 Review process	Offline	LO3		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
18	3.2 Review process (cont.)	Offline	LO3		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	3.2 Review process (cont.)	Offline	LO3		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
20	3.3 Static analysis by tools	Offline	LO3		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
21	Guiding Lab 3 (Static techniques: Manually review documents or use AI for static code analysis)	Offline	LO3, LO9		text book: Chapter 3 and Slides		Teamwork and Self Study - Read all relevant materials - Use Chat AI - Self Practice	
22	Topic 4. Test design techniques 4.1 The test development process	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
23	4.2 Categories of test design techniques	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
24	4.3 Specification-based or black-box techniques	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
25	4.3 Specification-based or black-box techniques (cont.)	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
26	4.3 Specification-based or black-box techniques (cont.)	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
27	4.3 Specification-based or black-box techniques (cont.)	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
28	4.4 Structure-based or white-box techniques	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
29	4.5 Experience-based techniques	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
30	4.6 Choosing a test technique	Offline	LO4		text book: Chapter 4 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
31	Progress test 2	Offline	LO3, LO4		text book: Chapter 3&4 and Slides		- Enter the exam system - Complete Progress Test	
32	Review Lab 2	Offline	LO2, LO9		text book: Chapter 2, AI Tool		Teamwork and Self Study - Read all materials in the module - Presentation & Review	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
33	Review Lab 3	Offline	LO3, LO9		text book: Chapter 3, AI Tool		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
34	Topic 5. Test management 5.1 Test organization	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
35	5.2 Test plans, estimates, and strategies	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
36	Guiding Lab 4 (Create Test Plan with AI)	Offline	LO5, LO9		text book: Chapter 5 and Tutorial		Teamwork and Self Study - Read all relevant materials - Use Chat AI - Self Practice	
37	5.3 Test progress monitoring and control	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
38	5.4 Configuration management	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
39	5.5 Risk and testing	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
40	5.6 Incident management	Offline	LO5, LO6		text book: Chapter 5 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
41	Review Lab 4	Offline	LO5, LO9		text book: Chapter 5, AI Tool		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
42	Review Problems	Offline	LO1 - LO9		Tutorial		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
43	Topic 6. Tool support for testing 6.1 Types of test tool	Offline	LO7		text book: Chapter 6 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
44	6.2 Effective use of tools: Potential benefits and risks	Offline	LO7		text book: Chapter 6 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
45	6.3 Introducing a tool into an organization	Offline	LO7		text book: Chapter 6 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
46	Progress Test 3	Offline	LO5, LO6, LO7		Tutorial		- Enter the exam system - Complete Progress Test	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
47	Topic 7. ISTQB Foundation Exam 7.1 Preparing for the exam 7.2 Taking the exam	Offline	LO1 - LO8		text book: Chapter 7 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
48	7.3 Mock exam	Offline	LO1 - LO8		text book: Chapter 7 and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
49	Topic 8. Agile Testing 8.1 The fundamentals of Agile software development 8.2 Aspects of Agile approaches	Offline	LO8		text book and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
50	8.3 Mostly used Agile Methods 8.4 Fundamental Agile Testing Principles, Practicles and Processes	Offline	LO8		text book and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
51	8.5 Agile Testing Methods, Techniques and Toools	Offline	LO8		text book and Slides		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
52	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
53	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
54	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
55	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
56	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
57	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
58	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
59	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	
60	Individual presentation and mark	Offline	LO2, LO4, LO5, LO6, LO7,LO10		Assignments submitted		Teamwork and Self Study - Read all materials in the module - Presentation & Review	

45 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	What challenges might arise when integrating AI into teaching software testing, and how can we address them?
2	2	CQ1.2	How would you explain the concept of "testing" to someone who has no background in software development?
3	3	CQ1.3	How does testing contribute to building trust in software systems for end-users?
4	4	CQ2.1	Why do you think defect clustering is a common phenomenon in software projects?
5	5	CQ2.2	How would you adapt the fundamental test process for a small project with limited resources?
6	6	CQ2.3	How can a tester's mindset differ from a developer's mindset, and why does this matter in testing?
7	7	CQ3.1	Why should ethical considerations, such as avoiding bias, be a priority in software testing?
8	8	CQ3.2	What are the advantages of integrating testing early in a V-Model compared to a Waterfall model?
9	9	CQ3.3	How does writing unit tests in an IDE help you understand the code's functionality?
10	10	CQ4.1	Why might integration testing reveal defects that were missed during component testing?
11	11	CQ4.2	How would you justify the need for acceptance testing to a project manager who wants to skip it?
12	12	CQ4.3	How would you decide whether to prioritize functional or non-functional testing for a mobile app?
13	13	CQ5.1	Why is regression testing critical after a software update, and how can it be optimized?
14	14	CQ5.2	How does maintenance testing differ from testing during initial development?
15	15	CQ5.3	What factors would you consider when planning maintenance testing for a legacy system?
16	16	CQ6.1	Why are reviews considered an effective way to find defects early in the development process?
17	17	CQ6.2	How does the role of a moderator impact the success of a review session?
18	18	CQ6.3	How would you handle a situation where team members disagree on a defect found during a review?
19	19	CQ7.1	Why is follow-up after a review as important as the review itself?
20	20	CQ7.2	What limitations of static analysis tools should you be aware of when using them?
21	21	CQ7.3	What criteria would you use to decide whether to use AI or manual methods for static analysis?
22	22	CQ8.1	Why is it important to define test conditions before designing test cases?
23	23	CQ8.2	When might you choose an experience-based technique over a specification-based one?
24	24	CQ8.3	How does equivalence partitioning reduce the number of test cases while maintaining coverage?
25	25	CQ9.1	Why is boundary value analysis particularly effective for finding defects?
26	26	CQ9.2	How can you use decision tables to ensure all possible scenarios are tested?
27	27	CQ9.3	How does state transition testing help in testing systems with complex workflows?
28	28	CQ10.1	Why might you combine multiple black-box techniques in a single test effort?
29	29	CQ10.2	Why might decision coverage be more valuable than statement coverage in critical systems?
30	30	CQ10.3	How can exploratory testing uncover defects that structured techniques might miss?
31	31	CQ11.1	What skills are most important for a tester using experience-based techniques?
32	32	CQ11.2	How would you decide which test technique to use for a time-constrained project?
33	33	CQ11.3	Why might the choice of test technique vary depending on the project's risk level?
34	34	CQ12.1	How does a clear test organization structure improve the efficiency of a testing team?
35	35	CQ12.2	Why is accurate test estimation critical to project success?
36	36	CQ12.3	How can metrics help you monitor testing progress effectively?
37	37	CQ13.1	What actions would you take if testing progress falls behind schedule?
38	38	CQ13.2	Why is configuration management essential for maintaining test integrity?
39	39	CQ13.3	How can risk analysis guide your testing efforts in a high-stakes project?
40	40	CQ14.1	What steps would you take to mitigate a newly identified risk during testing?

No.	Session No	Name	Details
41	41	CQ14.2	How would you prioritize incidents found during testing to ensure efficient resolution?
42	42	CQ14.3	Why is it important to document incidents thoroughly even if they are not fixed immediately?
43	43	CQ15.1	How do different types of test tools support various stages of the testing process?
44	44	CQ15.2	What risks might arise from over-relying on testing tools, and how can you mitigate them?
45	45	CQ15.3	Why is a pilot project important when introducing a new testing tool to an organization?

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Lab	on-going	4	25.0%	>0	Option 1: Lab 1: 2-6 weeks; Lab 2-4: 2 weeks. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO5, LO9, LO10	Option 1: Practical exercises using AI prompt; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: Lab1: Use chat AI to explore testing fundamentals through effective prompts; Lab2: Install and study an unit test framework, Use IDE to write unit test code and run unit test; Lab 3: Static techniques (Manually review documents or use AI for static code analysis); Lab 4: Create Test Plan with AI. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	The instructor support tutorials, students complete independently, submission by deadline	
2	Presentation	on-going	1	10.0%	>0	Option 1: 10'/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO2, LO4, LO5, LO6, LO7, LO10	Option 1: Personal Presentation; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A ;Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Test Plan, NUnit, JUnit, Test tool Selenium, Katalon Or Jira, Design test cases with Decision table testing, Use case testing techniques	Individual or team work, guided by instructor, submission by deadline	The ins recomm that ea apply t activiti their SV project team m respon a part c testing and pre individ about 8 minute The presen from sl 20.

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Progress Test	on-going	3	15.0%	>0	Option 1: 30'/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8	Option 1: Multiple choices Marked by Computer or a suitable format; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 20-30/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Test 1: Chapter 1 and 2; Test 2: Chapter 3 and 4; Test 3: Chapter 5 and 6	In class, by instructor	Instruc schedu Progre must b presen Course Implem Plan ap by dire the car Progre must b right af last lec require materia Instruc respon review for stu after gr
4	Final exam	Final exam	2	50.0%	4	145 minutes	All CLOs			All subjects in syllabus	By exam board Manual evaluating	Each p final ex be >=4 Final R >=5/10
5	Practical Exam	Final exam	1	25.0%	4	85'	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8	Assignment		Chapter 1 to 6	by Exam Board	Custon from th assign labs th studen learnec
6	Theory Exam	Final exam	1	25.0%	4	60'	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8	Multiple choicesMarked by Computer	60	Chapter 1 to 6	by exam board, using computer	The ex questio be upd differe least 7 the pre ones.